

Analysis Report: Phishing Detection and Awareness

By: Mahamat ADAM ADAM

Project: Future Interns – Task 2 (2026)

Table of contents

1. Introduction.....	1
2. My Analysis Methodology.....	1
Tools Used.....	1
3. Email Analysis and Classification.....	2
Analysis Summary Table	2
4. Detailed Phishing Indicators	2
A. Identity Spoofing	2
B. The SPF PASS Trap.....	3
C. Artificial Urgency	4
5. Survival Guide for Users (Prevention).....	5
✓ Best Practices.....	5
✗ What You Should Never Do.....	5
Conclusion.....	6

1. Introduction

Today, phishing is one of the most widespread threats in cybersecurity. Every day, millions of fraudulent emails are sent with the goal of deceiving users and stealing sensitive information such as passwords, personal data, or banking information.

Unlike many other cyberattacks, phishing does not directly target computer systems. Instead, it targets the human factor. Attackers exploit trust, curiosity, or fear to convince users to click on malicious links or disclose sensitive information.

As part of this project, my objective was to analyze several email samples in order to determine whether they are legitimate or malicious. To do this, I studied their technical headers, sender domains, and message content in order to identify common phishing indicators.

This analysis not only helps detect attacks but also raises awareness about the techniques used by cybercriminals.

2. My Analysis Methodology

To analyze the emails safely, I did not click on any links contained in the messages. Instead, I used several investigation tools and techniques to examine the technical elements of the emails.

Tools Used

Google Admin Toolbox – Message Header Analyzer

Link: <https://toolbox.googleapps.com/apps/messageheader/>

This tool allows the analysis of email headers, which contain important technical information such as:

- The real server that sent the email
- SPF, DKIM, and DMARC authentication results
- The path the email followed across the Internet
- Consistency between the sender domain and the sending server

Visual Content Analysis

- In addition to technical analysis, I also examined:
- The tone of the message
- Urgency or pressure tactics

- Spelling mistakes
- Suspicious links
- Unusual domains

These indicators often reveal phishing attempts.

3. Email Analysis and Classification

After analysis, I classified the emails according to three risk levels:

- 🟢 Safe (legitimate email)
- 🟡 Suspicious (potential risk)
- 🔴 Phishing (malicious email)

Analysis Summary Table

Source / Brand	Main Indicator	Technical Result	Risk	Classification
Leroy Merlin	Inconsistent sender	✗ SPF FAIL	Financial scam	Phishing
Microsoft	Fake security alert	⚠ DMARC ERROR	Account theft	Phishing
Amil Plan	Misleading marketing domain	✓ SPF PASS	Data collection	Suspicious
Official Notification	Verified identity	✓ SPF / DKIM / DMARC PASS	None	Safe

4. Detailed Phishing Indicators

During my investigation, several phishing indicators repeatedly appeared.

A. Identity Spoofing

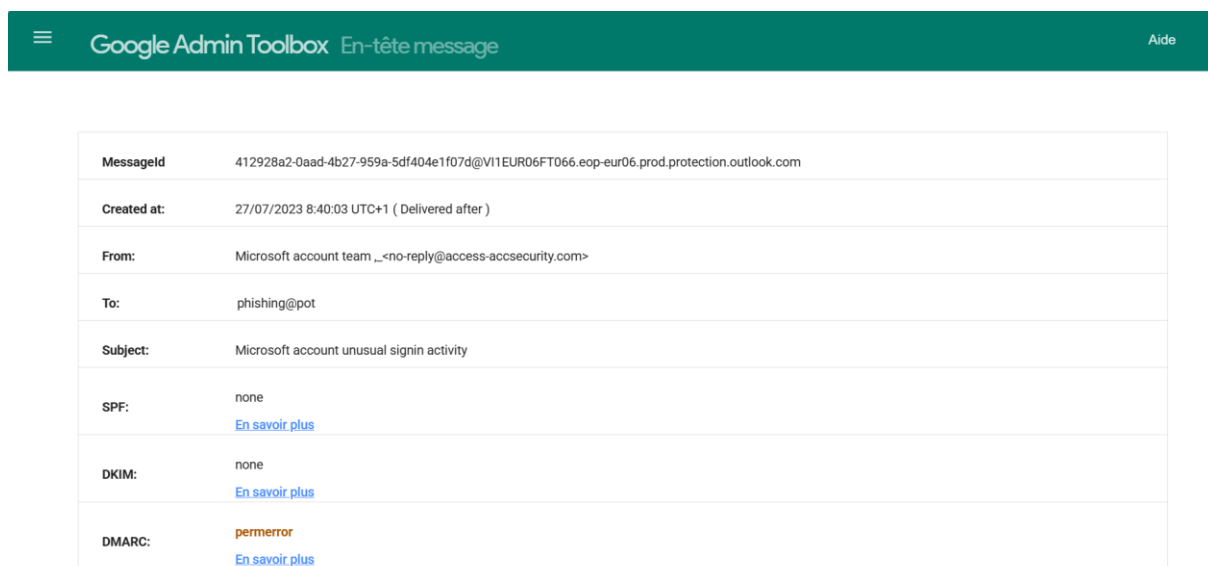
In the case of the email pretending to be from Microsoft, the displayed sender name was:

"Microsoft account team"

At first glance, this appears legitimate. However, after analyzing the email header using Google Toolbox, I discovered that the real address was:

no-reply@access-accsecurity.com

This domain does not belong to Microsoft. It is a fraudulent domain designed to imitate an official service.

The image shows a screenshot of the Google Admin Toolbox interface. At the top, there is a green header bar with a hamburger menu icon, the text "Google Admin Toolbox", "En-tête message", and a link labeled "Aide". Below this, a table displays the results of an email header analysis. The table has two columns: the field name and the value. The fields include MessageId, Created at, From, To, Subject, SPF, DKIM, and DMARC. The SPF, DKIM, and DMARC fields have additional links for further information.

MessageId	412928a2-0aad-4b27-959a-5df404e1f07d@V11EUR06FT066.eop-eur06.prod.protection.outlook.com
Created at:	27/07/2023 8:40:03 UTC+1 (Delivered after)
From:	Microsoft account team _<no-reply@access-accsecurity.com>
To:	phishing@pot
Subject:	Microsoft account unusual signin activity
SPF:	none En savoir plus
DKIM:	none En savoir plus
DMARC:	permerror En savoir plus

B. The SPF PASS Trap

The email related to the Amil health plan is particularly interesting because it shows a more sophisticated phishing case.

In Google Toolbox, the result indicates:

SPF : PASS

At first glance, this could reassure users. However, it does not necessarily mean that the email is legitimate.

An SPF PASS only indicates that the server used is authorized to send emails for that domain.

However, the domain used:

simuladordeplanodesaude.com

Does not belong to the official company. It is most likely a data collection platform or part of a phishing campaign.

Google Admin Toolbox En-tête message		Aide
MessageId	59549645-2f88-4bc5-b83d-ba483f8105b0@BN7NAM10FT091.eop-nam10.prod.protection.outlook.com	
Created at:	31/07/2023 10:19:56 UTC+1 (Delivered after 5 sec)	
From:	« Proposition de plans Amil » <marlene@a.simuladordeplanodesaude.com>	
To:	phishing@pot	
Subject:	Rodrigo, nous avons une proposition exclusive qui t'attend	
SPF:	pass avec l'adresse IP Unknown! En savoir plus	
DKIM:	none En savoir plus	
DMARC:	pass En savoir plus	

C. Artificial Urgency

Phishing emails often rely on a strategy known as social engineering.

In the Leroy Merlin example, the email promises a reward or surprise in order to encourage the user to click quickly on the malicious link.

This technique exploits:

- Curiosity
- Fear of missing out
- Time pressure

The goal is to prevent the user from thinking carefully or verifying the legitimacy of the message.

Google Admin Toolbox En-tête message		Aide
MessageId	SyvnJSL.0.0.SyvnJSL.9.SyvnJSL@ml.tv-news.fr	
Created at:	11/11/2023 15:45:52 UTC+1 (Delivered after 5 sec)	
From:	Message de Leroy Merlin, <programme_tv@ml.tv-news.fr>, Message de Leroy Merlin, <programme_tv@ml.tv-news.fr>	
To:	phishing@pot	
Subject:	Nous avons une surprise pour les clients de Leroy Merlin.	
SPF:	fail avec l'adresse IP Unknown! En savoir plus	
DKIM:	none En savoir plus	
DMARC:	none En savoir plus	

5. Survival Guide for Users (Prevention)

To reduce phishing risks, users should adopt a few simple security habits.

✓ Best Practices

Hover before clicking

Before clicking on any link, users should hover the mouse over it to check the real destination address.

If the address looks unusual or does not match the expected service, it is safer not to click.

Verify the real sender address

The displayed sender name can be manipulated. It is important to check the full sender email address.

Use analysis tools

When in doubt, email headers can be analyzed using tools like Google Admin Toolbox to verify message authenticity.

✗ What You Should Never Do

Never share your password via email.

No legitimate service (bank, online service, or email provider) will ask for your password through email.

Do not open unknown attachments.

Some attachments may contain malicious software.

Do not react under pressure.

Emails that demand immediate action are often part of phishing attacks.

Conclusion

This analysis shows that phishing attacks mainly rely on user manipulation.

Even when certain technical checks appear valid (such as the SPF PASS example), an email can still be malicious.

This is why user awareness remains one of the most effective ways to reduce cybersecurity risks.

By learning how to recognize phishing indicators and analyze suspicious emails, every user can become an important part of the organization's security defense.